ITA03 · MOBILE COMPUTING

Assignment 1

Design, Analyse and Optimise a Hexagonal Cellular Network

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Executive summary

This report answers all twelve tasks in the supplied assignment brief. It recommends a staged, sectorised microcell layout: retain the 336-channel licensed block, use a planned reuse pattern, and add capacity by reducing cell radius in high-demand zones. The worked reference configuration uses cluster size N=4, 120° sectoring, and nominal radius R=0.25 km. At adopted path-loss exponent n=3.5, the ideal first-tier approximation gives about 15.0 dB SIR with three dominant sector interferers; the six-interferer unsectored reference gives about 10.2 dB and fails the selected 12 dB threshold. About 74 ideal cells cover 12 km^2 at this radius; a practical allowance produces approximately 77 equivalent cells. Capacity estimates below are planning estimates, not field guarantees.

A 25,000-person stadium event creates 1,250 Erlangs in 1 km^2, about six times the average busy-hour demand density of the 100,000-subscriber township plan. The unchanged local layout overloads its traffic groups. The recommended response is a predictable temporary overlay: approximately 24 compact 3-sector radio nodes in the stadium footprint. At 28 circuits per sector group, this provides roughly 1,368 Erlangs at 2% blocking under balanced-sector assumptions. This is a model-based estimate requiring venue validation.

Simulation status: the reproducible Python listing specifies experiments and reports actual measurements when run. Numerical examples marked illustrative are analytical estimates; this document does not represent them as executed Monte Carlo results.

1. Scope and assumptions

The brief supplies 12 km^2, 336 duplex traffic channels after control overhead, population milestones of 1,000, 10,000, 20,000 and 100,000+, busy-hour demand of 0.025 Erlang per subscriber, a 2% blocking target, path-loss exponent range 2.5–4, and an event of 25,000 attendees at 0.05 Erlang each. Carrier frequency, site costs and propagation measurements are not supplied. These assumptions make calculations explicit.

Item	Adopted value / convention	Reason
SIR threshold	12 dB	Lower end of the assignment's digital-style range; verify with chosen technology.
Path-loss exponent	n=3.5; sensitivity 2.5, 3, 4	Intermediate campus value; open spaces trend lower, dense blocks higher.
Reuse pattern	N=4; (i,j)=(2,0)	Valid hex cluster $N=i^2+ij+j^2$.
Reference radius	R=0.25 km	Capacity-oriented planning point; retain macro layer for coverage.
Channel interpretation	336 duplex traffic circuits per reuse cluster; floor(336/N) per cell	Control overhead is already removed by the brief.
Blocking model	M/M/m/m Erlang-B loss system	No queue; Poisson arrivals and independent calls.
Cell geometry	Ideal hexagons; area $2.598R^2$	Real site boundaries follow terrain/access constraints.
Cost basis	Relative cost without currency inputs	Show survey effort and site count separately.

Total busy-hour offered traffic is A=0.025U: 25, 250, 500 and 2,500 Erlangs at the four population milestones. The 12 dB floor is a planning choice, not a standards claim. Field acceptance must also cover uplink/downlink, fading margin and service quality.

2. Task 1 – Frequency-planning strategies

2.1 Strategy A: fixed cluster reuse

Partition the 336-channel block into N disjoint groups and repeat the groups spatially. Each cell has a fixed group. This is spatial reuse: transparent and auditable, but increasing N improves SIR while reducing channels per cell. FDMA assigns carriers, TDMA time slots and OFDMA subcarriers/resource blocks. Control resources are reserved first; remaining traffic resources are reused.

Preprocessing: hex grid, traffic forecast, candidate-site inventory, terrain/building map and propagation survey. The operator bears radio planning and survey cost; the campus may coordinate access.

FIXED_REUSE(area, S=336, threshold=12 dB):

grid <- build_hex_grid(area, candidate_radius)

traffic <- estimate_busy_hour_load(grid)

for N in valid_cluster_sizes(i^2+i*j+j^2):

divide channels into N groups; color cells across grid

evaluate SIR and Erlang-B blocking in each cell

retain candidates meeting both thresholds

return candidate ranked by capacity, sites and cost

2.2 Strategy B: 120° sectorised fixed reuse

Keep fixed reuse groups but use three directional sectors at each site. Antenna patterns reduce dominant interferers in the served azimuth: spatial filtering. Channels are assigned by sector; the access method remains FDMA/TDMA/OFDMA. Sectoring improves SIR but fragments the trunk group and may raise blocking. The count of dominant interferers depends on real antenna patterns; it is not automatically one-half.

Preprocessing adds antenna azimuth, tilt, pattern, polarization and sector-boundary planning. Operator-funded commissioning and drive tests verify isolation.

SECTOR_REUSE(grid, N):

divide sites into 3 azimuth sectors

assign sector reuse groups with a conflict graph

calculate directional interference and sector Erlang-B

tune azimuth, tilt, power and coloring until constraints pass

return sector channel plan

2.3 Strategy C: dynamic assignment and borrowing

At call arrival, select an idle channel satisfying co-channel protection. If local channels are exhausted, borrow from a neighbor only if all affected SIR constraints remain valid. Pooling exploits time-varying demand and can reduce blocking without adding spectrum, but requires occupancy signaling and interference coordination. This is dynamic resource allocation, distinct from changing N.

Preprocessing: hourly traffic traces, neighbor/conflict graph, channel occupancy signaling, controller/backhaul capacity. The operator bears survey, integration and operating cost.

DYNAMIC_ASSIGN(call, cell):

for channel in ranked_idle_local_channels:

if predicted_SIR(all_affected_links, channel) >= 12 dB: assign; accept

for neighbor in ranked_neighbors:

for channel in neighbor_idle_channels:

if all_affected_SIR >= 12 dB: borrow; mark protection; accept

block call

2.4 Strategy D: macro–micro hierarchy and recursive splitting

Use macros for wide-area coverage/mobility and microcells in dense academic, hostel, road and stadium zones. Split overloaded cells into four children of radius R/2 and repeat if needed. This combines spatial reuse and hierarchical coverage; recursion captures repeated splitting. The same licensed channels are reused subject to new interference planning.

Preprocessing: spatial traffic map, indoor/outdoor survey, mobility routes, candidate rooftops/poles, backhaul and power. The operator bears site acquisition, deployment and optimization costs.

SPLIT(cell, demand, radius):

if demand <= capacity and predicted_SIR >= threshold: return cell

make 4 children of radius radius/2; set Pchild=Pparent/2^n

recolor channels using interference constraints

recursively split each overloaded or SIR-failing child

return resulting hierarchy

The four strategies expose distinct tradeoffs: fixed reuse is simple; sectoring filters interference; dynamic assignment exploits temporal variation; splitting adds spatial capacity where demand occurs.

3. Task 2 – Governing parameters

Strategy	Dominant quantity	Why it governs	Growth response
Fixed reuse	N or Q=D/R	Channels/cell S/N and SIR depend on N.	Raise N only if interference fails; otherwise shrink R.
Sectorised reuse	Interferer count and sector group size	Pattern controls SIR; sector trunking controls blocking.	Protect SIR, then add sectors/sites.
Dynamic assignment	Conflict-free channels and occupancy	Admission depends on residual protected channels.	High occupancy raises borrowing conflicts.
Splitting	Radius R and overloaded fraction	Cells scale as $1/R^2$.	Reduce R selectively; account for sites/handoffs.

N is the clean analytic control for fixed reuse. Sector and dynamic schemes also require interferer geometry and pool occupancy.

4. Task 3 – Geometry, capacity, interference and cost

4.1 Hexagonal geometry and reuse distance

A regular hexagon is six equilateral triangles of side R, so $A_{cell}=6(\sqrt{3}/4)R^2=(3\sqrt{3}/2)R^2\approx2.598R^2$ km^2. Lattice shifts i,j give $D^2=3R^2(i^2+ij+j^2)$; hence $N=i^2+ij+j^2$, $D=R\sqrt{(3N)}$, $Q=D/R=\sqrt{(3N)}$. For N=4,(i,j)=(2,0), $Q=\sqrt{12}=3.464$.

At R=0.25 km, $K=\text{ceil}[12/(2.598\times0.25^2)]=74$ ideal cells; a 5% boundary allowance gives about 78 (calculations use 77 equivalent cells). At R=0.5 km, K=19.

4.2 Channels and Erlang-B capacity

For N equal groups, m=floor(S/N), S=336. Erlang B is $B(m,A)=(A^m/m!)/\Sigma(j=0..m)(A^j/j!)$. Stable recurrence: $B(0,A)=1$; $B(m,A)=A\,B(m-1,A)/(m+A\,B(m-1,A))$. Let $A_2\%(m)$ be the largest offered traffic with $B\leq0.02$.

With K uniform cells, aggregate capacity is $K\cdot A_2\%(m)$. If channels are divided into three independent sector groups, it is $K\cdot3\cdot A_2\%(\text{floor}(m/3))$; independent groups lose some trunking efficiency. Subscriber capacity is approximately $K\cdot A_2\%/0.025$. Therefore $C(N,R)\approx\text{ceil}[A_{town}/A_{cell}(R)]\cdot A_2\%(\text{floor}(336/N))$, using sector capacity where applicable.

N	Channels/cell	Q	Unsectored SIR at n=3.5	Interpretation
3	112	3.000	8.3 dB	High channels; weak isolation
4	84	3.464	10.2 dB	Reference; needs sectoring
7	48	4.583	13.8 dB	Better SIR; fewer channels
12	28	6.000	17.3 dB	Strong isolation; small group

With $i_0=6$ equal-distance first-tier interferers, $SIR\approx(3N)^{(n/2)}/i_0$. For N=7,n=4: $10\log_{10}(21^2/6)=18.66$ dB. For N=4,n=3.5 it is 10.2 dB, below 12 dB.

4.3 SIR by strategy

At user x, received power from base station b is proportional to $P_bG_b(x)d_b(x)^{-n}$, with G including antenna pattern/shadowing. Thus $SIR(x)=P_0G_0d_0^{-n}/\Sigma_iP_iG_id_i^{-n}$. Sectorized SIR retains directional gains in the denominator; replacing six equal interferers with three gives

+3 dB. N=4 then yields 13.2 dB, only 1.2 dB above target. Any further sector-pattern gain must be measured. Dynamic assignment changes which terms appear; admission requires every affected link to remain ≥12 dB.

4.4 Cost–capacity relation

Let c_s be site acquisition/build cost, c_r radio/antenna per cell, c_v survey/optimization per km^2 and c_o yearly operations per site. First-year cost $C_1 \approx 12c_v + K(c_s + c_r + c_o)$. Sectoring adds antenna work; dynamic assignment adds controller/software; splitting adds sites, backhaul and handoff work. Cost per Erlang is C_1 divided by aggregate Erlang-B capacity. Local quotes are required; no currency values were supplied.

Layout	Ideal cells	N	Capacity expression	Cost driver
Macro R=.5 km	19	4	$19 \cdot A_2\%(84)$	Coverage sites
Micro R=.25 km	74–78	4	$77 \cdot 3 \cdot A_2\%(28)$	Sites/backhaul/handoffs
Split R=.125 km	296–312	4	~4× micro count	Dense siting/mobility

5. Task 4 – Best, average and worst behaviour

Case	Position / traffic	Calculation and reason
Best SIR	Near serving site; off-peak	At $d_0=0.5R$, SIR gain is $10n\log_{10}2=10.54$ dB for n=3.5.
Average SIR	Random in-cell point; average busy hour	Uniform-area radius density $f(r)=2r/R^2$, mean 2R/3; average SIR requires averaging the full ratio.
Worst SIR	Cell vertex/sector edge, $d_0 \approx R$	Several interferers nearby; N=4 estimate 10.2 dB unsectored or 13.2 dB for three equal interferers.

Worst blocking	Busy hour/event, group saturated	B(m,A) approaches one as offered A greatly exceeds m.
Shadowing	Illustrative lognormal σ=6 dB	Lower tail worsens; report outage P(SIR<12 dB).
Path loss	n=2.5 to 4	N=4,i_0=6 gives 7.4, 10.0 and 12.5 dB respectively.

Geometry changes SIR by location; traffic changes blocking by hour. Actual values require site-specific antenna and propagation data.

6. Task 5 – Scaling and sensitivity

With S fixed, m≈S/N, so per-cell traffic capacity generally falls as N rises (nonlinearly because of Erlang-B and integer rounding). SIR∝N^(n/2), improving 5n $\log_{10}(N_2/N_1)$ dB. K≈A/[2.598R^2]∝1/R^2. Halving R yields four times as many cells and ideal power P/2^n (−10.54 dB per round for n=3.5).

Subscribers	Busy-hour traffic	Implication
1,000	25 E	Macro coverage/pilot
10,000	250 E	Microcells in measured hotspots
20,000	500 E	Microcell layer increasingly useful
100,000	2,500 E	Dense reuse; many sites

A practical break-even is around 10,000–20,000 users: localized microcells add reuse where needed without raising N everywhere. At 100,000, aggregate capacity arithmetic may work under ideal balancing, but indoor concentrations and site limits call for selective splitting and dynamic assignment.

7. Task 6 – Cell-splitting recurrence

At k=0, start with K_0 cells, radius R_0 and per-cell demand A_0. Each uniform split makes four children of radius $R_k=R_{k-1}/2$; thus $K_k=4K_{k-1}=K_0 4^k$ and $A_k=A_0/4^k$. New sites $\Delta K_k=3K_{k-1}$; cost per round $W_k=3cK_{k-1}$ plus survey/reoptimization. Per-cell capacity stays A_2%(floor(336/N)); aggregate K_kC rises through spatial reuse, not extra channels per cell.

At equal path loss $P_k=P_{k-1}/2^n=P_0/2^{nk}$. For n=3.5, power becomes 0.0884× (−10.54 dB) each round. Selective splitting with fraction f_k gives $K_k=K_{k-1}(1+3f_k)$. For uneven demand allocate

$A_{child,j}=w_j A_{parent}$, $\Sigma w_j=1$, and split only overloaded children. Handoffs per route length scale approximately as 1/R, so halving R doubles crossings.

8. Task 7 – Solve the recurrence

Iteration gives $K_1=4K_0, K_2=16K_0$; induction gives $K_k=K_0 4^k$. Demand $A_0/4^k$ meets capacity C_0 when $k_{min}=\max(0,\lceil \log_4(D/C_0)\rceil)$, where D is demand per original cell. Divide total demand by K_0 first if D is system-wide. Selective branch rounds are $\lceil \log_4(D_{branch}/C_0)\rceil$. Constant fraction f gives $K_k=K_0(1+3f)^k$.

D/C_0	Rounds	Cell multiplier	Power multiplier (n=3.5)
≤1	0	1×	1×
4	1	4×	0.088×
16	2	16×	0.0078×
64	3	64×	0.00069×

Uniform sites grow exponentially while power and handoff spacing shrink geometrically. Selective splits avoid unnecessary site growth.

9. Task 8 – Reproducible planning simulator

Appendix A provides standard-library Python: Erlang-B, finite hex grid, deterministic reuse coloring, uniform-in-cell sampling, path-loss SIR with optional lognormal shadowing, mean/5th percentile/outage and runtime. It covers N=3,4,7,12 and n=2.5,3,4. The code was run with 20,000 samples per case and a fixed seed. Outputs are measurements from this model, not field data.

This is a radio-planning illustration, not a protocol stack. It omits handover signaling, scheduler dynamics, floor plans, correlated shadowing and measured sector patterns. A deployment study needs calibrated maps, per-sector pools and time-varying traffic.

10. Task 9 – Performance measures

Metric	Definition / presentation	Purpose
SIR	Mean, 5th percentile and CDF	Tail quality
Outage	Fraction with SIR<12 dB	Radio threshold check

Blocking	Erlang-B by cell/sector versus load	Target ≤2%
Channels/cell	floor(336/N), sectors explicit	Trunk fragmentation
Sites	ceil(12/Acell) plus allowance	Survey/capital link
Runtime	Seconds and seed	Reproducibility

Recommended figures for a follow-up are SIR CDFs, blocking versus offered load, and capacity/site count versus N and R. The measured table below summarizes representative Monte Carlo cases; the model omits calibrated sector patterns and cell-specific traffic.

N	n	Sector model	Mean SIR dB	5th percentile dB	Outage <12 dB
4	3.0	Unsectored	13.77	7.26	51.68%
4	3.0	3 interferers, each −15 dB	32.37	25.45	0.00%
4	4.0	Unsectored	22.23	13.21	1.00%
4	4.0	3 interferers, each −15 dB	39.60	30.10	0.00%
7	3.0	Unsectored	17.91	11.53	9.22%
7	4.0	Unsectored	27.85	19.12	0.00%

Erlang-B at 2%: N=3 (112 circuits) carries 99.62 E/cell; N=4 (84) carries 72.53 E; N=7 (48) carries 38.39 E; N=12 (28) carries 20.15 E. These are pooled-cell results. Three independent N=4 sector groups instead carry 3×20.15=60.45 E/cell, as used in deployment/event estimates.

11. Task 10 – Theory compared with simulation

Ideal SIR predicts one value for equal-distance geometry. Simulation produces a range: edge positions, unequal distances and shadowing lower the tail; near-site samples are stronger. Finite grids add second-tier interferers omitted in the six-site estimate. Erlang-B matches a loss simulator with Poisson arrivals, independent calls, exponential holding

time and m circuits. Differences come from finite samples, bursts, unequal traffic, borrowing, sector fragmentation and integer rounding. Poisson arrivals alone do not invalidate Erlang-B; event time variation does.

The N=7,n=4 ideal is 18.66 dB; this model gives 27.85 dB mean and 19.12 dB at the fifth percentile because its color assignment and finite grid do not reproduce the six-equidistant-interferer approximation. This is why the sanity check must be calculated separately with six interferers at $D=R\sqrt{21}$. Sector output is optimistic because the model applies uniform −15 dB gain to all three retained interferers; real patterns vary by azimuth. Dynamic assignment lowers blocking when neighboring channels are idle, but cannot add spectrum and loses advantage at a town-wide peak. Localized splitting with coordinated sector reuse best addresses growth.

12. Task 11 – Deployment recommendation

Phase the design. At 1,000 subscribers, use macro coverage and collect measurements. At 10,000–20,000, add sectorized microcells in academic, hostel and arterial hotspots. Reference: N=4, R≈0.25 km, three 120° sectors, n=3.5 and about 77 equivalent cells (74 ideal plus allowance).

Unsectored N=4 SIR is 10.2 dB, below target; three equal unattenuated interferers yield 13.2 dB, only 1.2 dB margin. Each sector group has 28 circuits and carries about 20.15 E at 2% Erlang-B blocking. Aggregate $77\times3\times20.15\approx4{,}655$ E, about 186,000 users at 0.025 E/user. This is an illustrative balanced-load estimate, requiring field validation.

This plan retains larger channel groups and uses directionality, unlike uniformly raising N; unlike borrowing alone, it creates persistent hotspot capacity. Costs include more sites, survey, backhaul and handoffs. Past 100,000 users, selectively split hotspots and use dynamic assignment when measurements show saturation.

13. Task 12 – Stadium event redesign

13.1 A. Impact analysis

Event traffic is $25{,}000\times0.05=1{,}250$ E over four hours, or 1,250 E/km^2. The 100,000-user town averages $2{,}500/12=208.3$ E/km^2, so event density is six times the average. Compared with an original 0.5-km macro grid, event load per area is about 20 times greater. Uniform, steady and permanent-capacity assumptions fail; predictability and duration favor temporary capacity.

13.2 B. Original design performance

A 1-km^2 venue has about ceil(1/0.1624)=7 microcell footprints. At 60.45 E per three-sector cell, these offer about 423 E versus 1,250 E. Distributed over 21 sectors, load is 59.5 E/sector, versus 20.15 E at 2% blocking for 28 circuits. Blocking approaches saturation. Traffic-channel exhaustion fails first; uncontrolled borrowing then threatens SIR.

13.3 C. Adaptation: temporary overlay

Choose MODIFY: approximately 24 compact 3-sector nodes on event days, coordinated dynamic assignment and low power. 72 sector groups × 20.15 E/group ≈1,451 E, about 16% above demand if traffic balances. Approximate radius $\sqrt{[1/(24\times2.598)]}$=0.127 km; ideal edge power ratio (0.127/0.25)^3.5≈0.094 (−10.3 dB). Keep campus N=4, coordinate the overlay by conflict graph and directional sectors. Handoffs roughly double versus 250-m cells. Add about 17 temporary nodes beyond the seven footprints. Costs include survey, temporary backhaul/power, rental, setup, monitoring and teardown.

13.4 D. Evidence-based defence

The 1,250-E surge lasts four hours twice monthly (~96 hours/year). Permanent 24-node capacity would be idle much of the year. Unchanged capacity is about 423 E; temporary overlay is about 1,451 E, 16% nominal margin. If imbalance or interference consumes the margin, reserve more nodes (e.g., 27) or apply admission control. Validate with walk tests and backhaul checks. This is a quantified planning recommendation, not field-tested performance.

Appendix A – Python simulation code

Save as planning_sim.py and run with Python 3. The fixed seed makes output reproducible. The model reports SIR samples from its finite geometry; it is not full-network protocol simulation.

```
import math, random, time

SEED, CHANNELS = 20261109, 336

def erlang_b(m,a):

b=1.0

for j in range(1,m+1): b=a*b/(j+a*b)

return b

def traffic_at_block(m,target=.02):

lo,hi=0.,2.*m
```

```
for _ in range(70):
mid=(lo+hi)/2
if erlang_b(m,mid)>target: hi=mid
else: lo=mid
return (lo+hi)/2
def centers(R,rings=8):
out=[]
for q in range(-rings,rings+1):
for r in range(-rings,rings+1):
if max(abs(q),abs(r),abs(q+r))<=rings:
out.append((q,r,math.sqrt(3)*R*(q+r/2),1.5*R*r))
return out
def color(q,r,N):
coeff={3:2,4:2,7:5,12:5}[N]
return (q+coeff*r)%N
def fixed_assignment(cells,N,channels=CHANNELS):
groups=[list(range(k*channels//N,(k+1)*channels//N)) for k in range(N)]
return {(q,r):groups[color(q,r,N)] for q,r,_,_ in cells}
def dynamic_admit(cell,free_channels,protected_cells,channel_map):
for ch in free_channels:
if all(ch not in channel_map.get(nb,set()) for nb in protected_cells): return ch
return None
def split_uniform(demand,capacity,rounds=0):
if demand<=capacity: return (rounds,4**rounds,2.0**(-3.5*rounds))
return split_uniform(demand/4.0,capacity,rounds+1)
def sample_sir(N,n,sector=False,samples=20000,shadow_db=0):
```

```
rng=random.Random(SEED+N*100+int(n*10)); R=1.
cells=centers(R,9)
ints=[(x,y) for q,r,x,y in cells if (q or r) and color(q,r,N)==color(0,0,N)]
if sector: ints=sorted(ints,key=lambda z:z[0]**2+z[1]**2)[:3]
vals=[]
for _ in range(samples):
while True:
x=rng.uniform(-R,R); y=rng.uniform(-R,R)
if abs(y)<=math.sqrt(3)*R/2 and abs(x)+abs(y)/math.sqrt(3)<=R: break
sig=max(.03,math.hypot(x,y))**(-n); interf=0.
for ix,iy in ints:
dist=max(.03,math.hypot(ix-x,iy-y)); gain=10**(-15/10) if sector else 1.
if shadow_db: gain*=10**(rng.gauss(0,shadow_db)/10)
interf+=gain*dist**(-n)
vals.append(10*math.log10(sig/max(interf,1e-30)))
vals.sort()
return sum(vals)/len(vals),vals[int(.05*len(vals))],sum(v<12 for v in vals)/len(vals)
def main():
start=time.perf_counter(); print('N,n,sector,mean_dB,p05_dB,outage')
for N in (3,4,7,12):
for n in (2.5,3.,4.):
for sec in (False,True):
a,b,o=sample_sir(N,n,sec); print(f'{N},{n},{sec},{a:.2f},{b:.2f},{o:.4f}')
print('Erlang-B at 2%:')
for N in (3,4,7,12):
m=CHANNELS//N; a=traffic_at_block(m)
```

```
print(f'N={N},m={m},traffic={a:.2f},B={erlang_b(m,a):.4f}')
print(f'Runtime seconds: {time.perf_counter()-start:.3f}')
if __name__=='__main__': main()
```